# Newton's Method

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> **Source**
>
> This section is adapted from the Newton-Raphson and gradient descent parts of Zhou [1], used under the MIT License (see the license text in Linear Algebra). The source motivates the Newton step informally, as the step that minimizes a quadratic approximation; here Theorem 0.1 proves that the step is that approximation's exact minimizer. In the source's zigzag example the starting value of the objective is $\frac{1}{2}(b^2+b)$, not $\frac{1}{2}(1+b^2)$ as the source writes; the ratio of successive values is unaffected. In the same example the source's formula for $\underset{\sim}{x}^{(1)}$ has $\underset{\sim}{x}^{(1)}$ on its right side where $\underset{\sim}{x}^{(0)}$ is meant. These parts of the source are not part of this section:
>
> - the quadratic convergence theorem for Newton's method in general, and its remedies for instability (a positive definite substitute for the Hessian, and shortening the Newton step by a search along its direction)
> - its remark that Hessians are expensive to compute
> - the convergence-rate theorem for strongly convex functions
> - gradient descent with momentum, and its optimal parameters
> - Nesterov acceleration, and stochastic gradient descent
> - its code and animations

## Iterative methods

**Definition 0.1** (Iterative method)**.** An **iterative method** [2] starts from a point $\underset{\sim}{x}^{(0)}$ and computes $\underset{\sim}{x}^{(t+1)}$ from $\underset{\sim}{x}^{(t)}$ by a fixed rule, for $t = 0, 1, 2, \ldots$.

**Definition 0.2** (Iterate)**.** In an iterative method (Definition 0.1), the points $\underset{\sim}{x}^{(t)}$ are its **iterates**.

**Definition 0.3** (Error of an iterate)**.** For a point $\underset{\sim}{x}^*$ that an iterative method (Definition 0.1) is meant to reach, $\|\underset{\sim}{x}^{(t)} - \underset{\sim}{x}^*\|$ is the **error** of the iterate $\underset{\sim}{x}^{(t)}$ (Definition 0.2).

**Definition 0.4** (Convergence of a sequence of vectors)**.** A sequence $\underset{\sim}{x}^{(0)}, \underset{\sim}{x}^{(1)}, \ldots$ in $\mathbb{R}^p$ **converges** to $x^* \in \mathbb{R}^p$ if the sequence of distances $\|\underset{\sim}{x}^{(t)} - \underset{\sim}{x}^*\|$ converges to 0. For $p = 1$, $\|x^{(t)} - x^*\| = |x^{(t)} - x^*|$, and this is the convergence of a sequence of numbers to $x^*$.

**Example 0.1** (Halving a vector)**.** The iterative method $\underset{\sim}{x}^{(t+1)} = \frac{1}{2}\underset{\sim}{x}^{(t)}$ from $\underset{\sim}{x}^{(0)} = (4,2)^\top$ has iterates $(4,2)^\top, (2,1)^\top, (1,0.5)^\top, \ldots$, that is, $\underset{\sim}{x}^{(t)} = 2^{-t}(4,2)^\top$. With $\underset{\sim}{x}^* = \underset{\sim}{0}$, the error of $\underset{\sim}{x}^{(t)}$ is

$$\begin{aligned} \left\|2^{-t}(4,2)^\top - \underset{\sim}{0}\right\| &= 2^{-t}\left\|(4,2)^\top\right\| && \text{(norm of a multiple)} \\ &= 2^{-t}\sqrt{20}, && (4^2 + 2^2 = 20) \end{aligned}$$

which converges to 0, so the iterates converge to $\underset{\sim}{0}$.

## Newton's method and the quadratic approximation

**Definition 0.1** (Newton's method for minimization)**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ have first and second partial derivatives, and let $\underset{\sim}{x}^{(0)}$ be a starting point. **Newton's method** [3] computes, for $t = 0, 1, 2, \ldots$, as long as $\mathbf{H}_f(\underset{\sim}{x}^{(t)})$ is invertible,

$$\underset{\sim}{x}^{(t+1)} \overset{\text{def}}{=} \underset{\sim}{x}^{(t)} - \left[\mathbf{H}_f(\underset{\sim}{x}^{(t)})\right]^{-1} \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^{(t)}).$$

**Definition 0.2** (Newton step)**.** In Newton's method (Definition 0.1), the vector $-\left[\mathbf{H}_f(\underset{\sim}{x}^{(t)})\right]^{-1} \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^{(t)})$ is the **Newton step** at $\underset{\sim}{x}^{(t)}$.

**Example 0.1** (Newton's method for a cubic)**.** Let $f(x) = \frac{1}{3}x^3 - 4x$, so $f'(x) = x^2 - 4$ and $f''(x) = 2x$. For $x^{(t)} \neq 0$, one Newton step gives

$$
\begin{aligned}
x^{(t+1)} &= x^{(t)} - \frac{(x^{(t)})^2 - 4}{2x^{(t)}} && \text{(Newton's method with } p = 1) \\
&= \frac{2(x^{(t)})^2 - (x^{(t)})^2 + 4}{2x^{(t)}} && \text{(common denominator)} \\
&= \frac{1}{2}\left(x^{(t)} + \frac{4}{x^{(t)}}\right). && \text{(simplify)}
\end{aligned}
$$

From $x^{(0)} = 2.5$:

- $$
\begin{aligned}
x^{(1)} &= \tfrac{1}{2}(2.5 + 1.6) \\
&= 2.05;
\end{aligned}
$$
- $$
\begin{aligned}
x^{(2)} &= \tfrac{1}{2}(2.05 + 1.951219\ldots) \\
&= 2.000610
\end{aligned}
$$

  (to six decimals);
- $x^{(3)} = 2.0000001$ (to seven decimals).

The iterates (Definition 0.2) approach 2, the local minimizer of $f$ (a stationary point, since $f'(2) = 0$, with $f''(2) = 4 > 0$; Theorem 7 in Critical Points and Optimality Conditions).

**Definition 0.3** (Quadratic approximation, quadratic model, second-order Taylor polynomial)**.** Let $f$ be as in Definition 0.1 and let $\underset{\sim}{x}$ be a point, with $\underset{\sim}{g} \stackrel{\text{def}}{=} \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x})$ and $\mathbf{H} \stackrel{\text{def}}{=} \mathbf{H}_f(\underset{\sim}{x})$. The **quadratic approximation** of $f$ at $\underset{\sim}{x}$ is the function of a step $\underset{\sim}{\delta} \in \mathbb{R}^p$

$$
q(\underset{\sim}{\delta}) \stackrel{\text{def}}{=} f(\underset{\sim}{x}) + \underset{\sim}{g} \cdot \underset{\sim}{\delta} + \tfrac{1}{2}\, \underset{\sim}{\delta}^{\top} \mathbf{H} \underset{\sim}{\delta},
$$

also called the **quadratic model** or the **second-order Taylor polynomial** of $f$ at $\underset{\sim}{x}$.

**Example 0.2** (The quadratic approximation at $x = 2.5$)**.** For $f(x) = \frac{1}{3}x^3 - 4x$ at $x = 2.5$ (Example 0.1),

$$
\begin{aligned}
g &= 2.5^2 - 4 \\
&= 2.25
\end{aligned}
$$

and

$$H = [2 \cdot 2.5]$$
$$= [5],$$

so $q(\delta) = f(2.5) + 2.25\,\delta + 2.5\,\delta^2$. For $f(\underset{\sim}{x}) = \frac{1}{2}(x_1^2 + 3x_2^2)$, whose Hessian is $\begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$ at every point, the quadratic approximation at any $\underset{\sim}{x}$ equals $f(\underset{\sim}{x} + \underset{\sim}{\delta})$ exactly (Example 12 in Critical Points and Optimality Conditions with $\mathbf{S} = \begin{bmatrix} 1/2 & 0 \\ 0 & 3/2 \end{bmatrix}$).

**i** Remark (The quadratic approximation and Taylor's theorem)

*Remark* 0.1 (The quadratic approximation and Taylor's theorem). When the second partial derivatives of $f$ are continuous, Theorem 6 in Critical Points and Optimality Conditions says that $f(\underset{\sim}{x} + \underset{\sim}{\delta})$ equals the expression for $q(\underset{\sim}{\delta})$ in Definition 0.3 with the Hessian evaluated at a point between $\underset{\sim}{x}$ and $\underset{\sim}{x} + \underset{\sim}{\delta}$.

**Theorem 0.1** (The Newton step minimizes the quadratic approximation)**.** *Let $q$ be the quadratic approximation of $f$ at $\underset{\sim}{x}$, with $\mathbf{H} = \mathbf{H}_f(\underset{\sim}{x})$ positive definite. Its unique minimizer is the Newton step $\underset{\sim}{\delta}^* = -\mathbf{H}^{-1}\underset{\sim}{g}$.*

**i** Proof

*Proof.* $\mathbf{H}$ is symmetric, by the definition of positive definite, and invertible, so $\underset{\sim}{\delta}^*$ exists and $\mathbf{H}\underset{\sim}{\delta}^* = -\underset{\sim}{g}$. For any $\underset{\sim}{d} \in \mathbb{R}^p$, first

$$\begin{aligned} \underset{\sim}{d}^\top \mathbf{H} \underset{\sim}{\delta}^* &= -\underset{\sim}{d} \cdot \underset{\sim}{g} && (\mathbf{H}\underset{\sim}{\delta}^* = -\underset{\sim}{g}) \\ &= -\underset{\sim}{g} \cdot \underset{\sim}{d}, && (\text{a } 1 \times 1 \text{ matrix equals its transpose}) \end{aligned}$$

and ${\underset{\sim}{\delta}^*}^\top \mathbf{H} \underset{\sim}{d} = \underset{\sim}{d}^\top \mathbf{H} \underset{\sim}{\delta}^*$ in the same way, using $\mathbf{H}^\top = \mathbf{H}$. Then

$$\begin{aligned} q(\underset{\sim}{\delta}^* + \underset{\sim}{d}) &= f(\underset{\sim}{x}) + \underset{\sim}{g} \cdot (\underset{\sim}{\delta}^* + \underset{\sim}{d}) + \tfrac{1}{2}\,(\underset{\sim}{\delta}^* + \underset{\sim}{d})^\top \mathbf{H} (\underset{\sim}{\delta}^* + \underset{\sim}{d}) && (\text{definition of } q) \\ &= f(\underset{\sim}{x}) + \underset{\sim}{g} \cdot \underset{\sim}{\delta}^* + \underset{\sim}{g} \cdot \underset{\sim}{d} + \tfrac{1}{2}\left({\underset{\sim}{\delta}^*}^\top \mathbf{H} \underset{\sim}{\delta}^* + {\underset{\sim}{\delta}^*}^\top \mathbf{H} \underset{\sim}{d} + \underset{\sim}{d}^\top \mathbf{H} \underset{\sim}{\delta}^* + \underset{\sim}{d}^\top \mathbf{H} \underset{\sim}{d}\right) && (\text{distribute}) \\ &= f(\underset{\sim}{x}) + \underset{\sim}{g} \cdot \underset{\sim}{\delta}^* + \underset{\sim}{g} \cdot \underset{\sim}{d} + \tfrac{1}{2}\,{\underset{\sim}{\delta}^*}^\top \mathbf{H} \underset{\sim}{\delta}^* - \underset{\sim}{g} \cdot \underset{\sim}{d} + \tfrac{1}{2}\,\underset{\sim}{d}^\top \mathbf{H} \underset{\sim}{d} && (\text{both cross terms equal } -\underset{\sim}{g} \cdot \underset{\sim}{d}) \\ &= f(\underset{\sim}{x}) + \underset{\sim}{g} \cdot \underset{\sim}{\delta}^* + \tfrac{1}{2}\,{\underset{\sim}{\delta}^*}^\top \mathbf{H} \underset{\sim}{\delta}^* + \tfrac{1}{2}\,\underset{\sim}{d}^\top \mathbf{H} \underset{\sim}{d} && (\text{cancel } \underset{\sim}{g} \cdot \underset{\sim}{d}) \\ &= q(\underset{\sim}{\delta}^*) + \tfrac{1}{2}\,\underset{\sim}{d}^\top \mathbf{H} \underset{\sim}{d}, && (\text{definition of } q) \end{aligned}$$

which is greater than $q(\underset{\sim}{\delta}^*)$ for every $\underset{\sim}{d} \neq \underset{\sim}{0}_{p\times 1}$, since $\mathbf{H}$ is positive definite. □

**Example 0.3** (The Newton step at $x = 2.5$)**.** For the quadratic approximation $q(\delta) = f(2.5) + 2.25\,\delta + 2.5\,\delta^2$ of Example 0.2, $H = [5]$ is positive definite, so by Theorem 0.1 its minimizer is

$$\begin{aligned}\delta^* &= -H^{-1}g \\ &= -2.25/5 \\ &= -0.45.\end{aligned}$$

Setting the derivative $2.25 + 5\delta$ to 0 gives the same $\delta^*$, and

$$\begin{aligned}2.5 - 0.45 &= 2.05 \\ &= x^{(1)}\end{aligned}$$

(Example 0.1).

## The Newton step as a root of the linear approximation of the gradient

**Definition 0.1** (Linear approximation of the gradient)**.** Let $f$ be as in Newton's method and let $\underset{\sim}{x}$ be a point. The **linear approximation of the gradient** of $f$ at $\underset{\sim}{x}$ is the $p \times 1$ vector-valued function $\underset{\sim}{g}_q$ of a step $\underset{\sim}{\delta} \in \mathbb{R}^p$ whose $j$th entry, for $j = 1, \ldots, p$, is the function whose graph is the tangent plane to the graph of the partial derivative $\frac{\partial}{\partial x_j} f$ at $\underset{\sim}{x}$, evaluated at $\underset{\sim}{x} + \underset{\sim}{\delta}$:

$$\left[\underset{\sim}{g}_q(\underset{\sim}{\delta})\right]_j \stackrel{\text{def}}{=} \frac{\partial}{\partial x_j} f(\underset{\sim}{x}) + \left(\frac{\partial}{\partial \underset{\sim}{x}}\left(\frac{\partial}{\partial x_j} f(\underset{\sim}{x})\right)\right)^{\top} \underset{\sim}{\delta}. \tag{1}$$

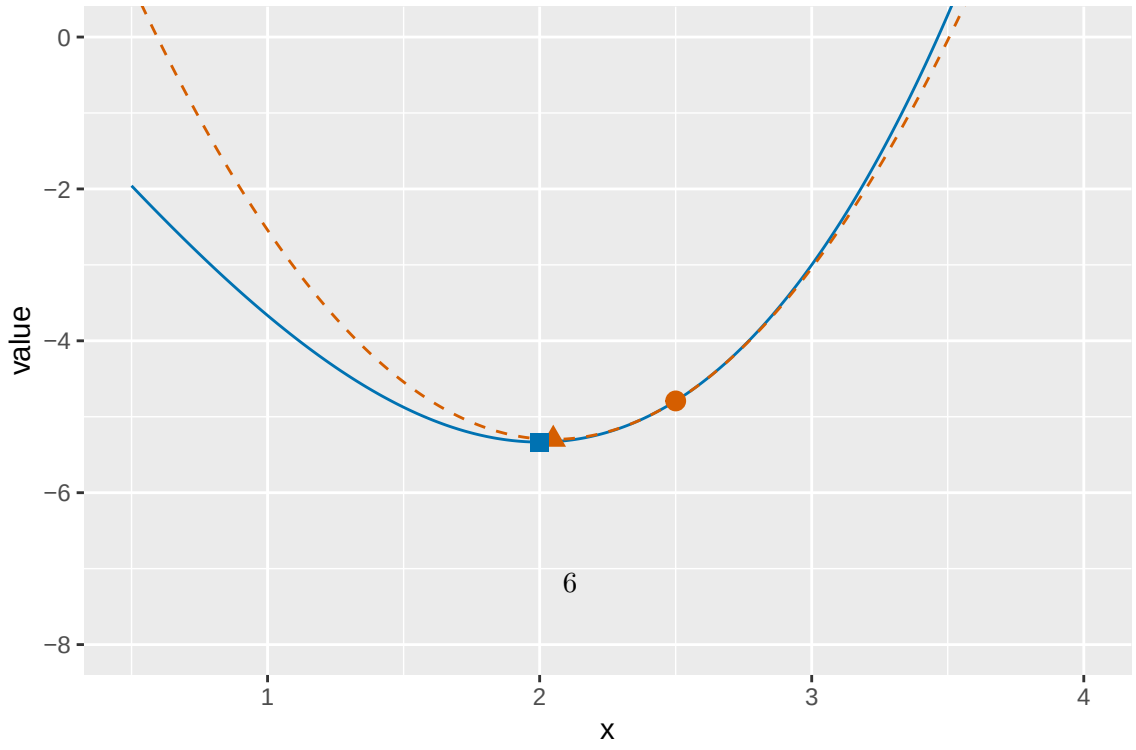
**Example 0.1** (The linear approximation of the derivative of a cubic at $x = -2.5$)**.** For $f(x) = \frac{1}{3}x^3 - 4x$, with $f'(x) = x^2 - 4$ and $f''(x) = 2x$ (Example 0.1), $p = 1$, so the linear approximation of the gradient at $x = -2.5$ is the tangent line to $f'$ there:

$$\begin{aligned}\ell(\delta) &= f'(-2.5) + f''(-2.5)\,\delta && \text{(Equation 1 with } p = 1) \\ &= ((-2.5)^2 - 4) + 2(-2.5)\,\delta && \text{(substitute } f' \text{ and } f'') \\ &= 2.25 - 5\delta. && \text{(arithmetic)}\end{aligned} \tag{2}$$

Figure 2 shows $f'$ and $\ell$.

**R**

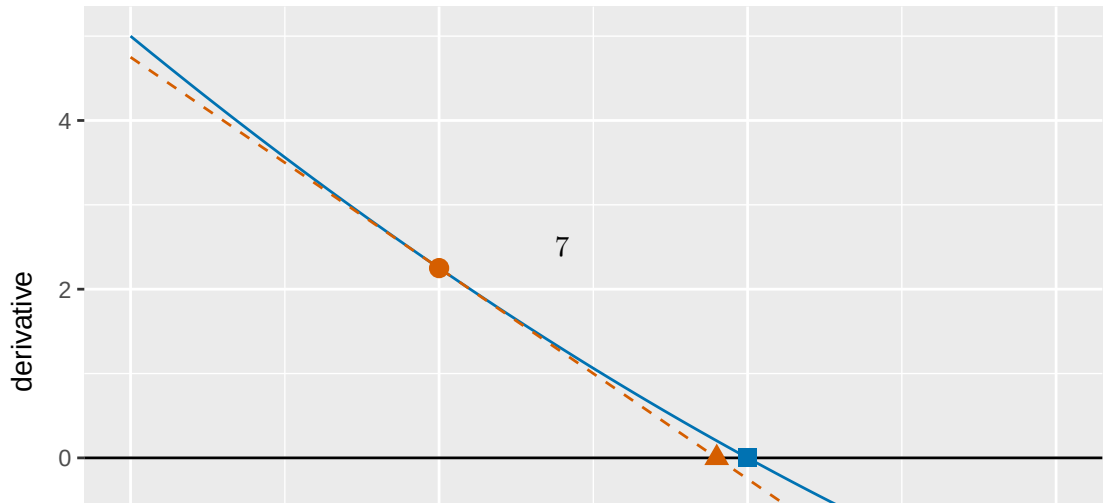
```r
nm_x_nm <- seq(0.5, 4, length.out = 301)
nm_f_nm <- nm_x_nm^3 / 3 - 4 * nm_x_nm
nm_f0_nm <- 2.5^3 / 3 - 4 * 2.5
nm_q_nm <- nm_f0_nm + 2.25 * (nm_x_nm - 2.5) + 2.5 * (nm_x_nm - 2.5)^2
nm_q_step_nm <- nm_f0_nm + 2.25 * (2.05 - 2.5) + 2.5 * (2.05 - 2.5)^2
nm_f_min_nm <- 2^3 / 3 - 4 * 2
nm_curve_nm <- tibble::tibble(x = nm_x_nm, y = nm_f_nm)
nm_model_nm <- tibble::tibble(x = nm_x_nm, y = nm_q_nm)
nm_start_nm <- tibble::tibble(x = 2.5, y = nm_f0_nm)
nm_step_nm <- tibble::tibble(x = 2.05, y = nm_q_step_nm)
nm_min_nm <- tibble::tibble(x = 2, y = nm_f_min_nm)
nm_aes_nm <- ggplot2::aes(x, y)
nm_ylim_nm <- c(-8, 0)
ggplot2::ggplot(nm_curve_nm, nm_aes_nm) +
  ggplot2::geom_line(colour = "#0072B2") +
  ggplot2::geom_line(
    data = nm_model_nm, colour = "#D55E00", linetype = "dashed"
  ) +
  ggplot2::geom_point(data = nm_start_nm, colour = "#D55E00", size = 3) +
  ggplot2::geom_point(
    data = nm_step_nm, colour = "#D55E00", size = 3, shape = 17
  ) +
  ggplot2::geom_point(
    data = nm_min_nm, colour = "#0072B2", size = 3, shape = 15
  ) +
  ggplot2::coord_cartesian(ylim = nm_ylim_nm) +
  ggplot2::labs(x = "x", y = "value")
```




**Python**

**R**

```
gla_x <- seq(-3, -1.5, length.out = 301)
gla_deriv <- gla_x^2 - 4
gla_line <- 2.25 - 5 * (gla_x + 2.5)
gla_curve <- tibble::tibble(x = gla_x, y = gla_deriv)
gla_tangent <- tibble::tibble(x = gla_x, y = gla_line)
gla_start <- tibble::tibble(x = -2.5, y = 2.25)
gla_step <- tibble::tibble(x = -2.05, y = 0)
gla_root <- tibble::tibble(x = -2, y = 0)
gla_ylim <- c(-2, 5)
gla_curve |>
  ggplot2::ggplot() +
  ggplot2::aes(x = x, y = y) +
  ggplot2::geom_hline(yintercept = 0) +
  ggplot2::geom_line(colour = "#0072B2") +
  ggplot2::geom_line(
    data = gla_tangent, colour = "#D55E00", linetype = "dashed"
  ) +
  ggplot2::geom_point(data = gla_start, colour = "#D55E00", size = 3) +
  ggplot2::geom_point(
    data = gla_step, colour = "#D55E00", size = 3, shape = 17
  ) +
  ggplot2::geom_point(
    data = gla_root, colour = "#0072B2", size = 3, shape = 15
  ) +
  ggplot2::coord_cartesian(ylim = gla_ylim) +
  ggplot2::labs(x = "x", y = "derivative")
```



**Example 0.2** (The tangent line to $f$ is not the linear approximation of the gradient)**.** For the cubic $f(x) = \frac{1}{3}x^3 - 4x$ at $x = -2.5$ (Example 0.1), the tangent line to $f$ itself is

$$\begin{aligned} f(-2.5) + f'(-2.5)\,\delta &= \left(\tfrac{1}{3}(-2.5)^3 - 4(-2.5)\right) + \left((-2.5)^2 - 4\right)\delta && \text{(substitute } f \text{ and } f'\text{)} \\ &= \tfrac{115}{24} + 2.25\,\delta. && \text{(arithmetic)} \end{aligned} \tag{3}$$

It is not the linear approximation of the gradient: it approximates $f$, not $f'$. Its root, $\delta = -\frac{115}{54}$, is a step toward a zero of $f$, not toward a stationary point of $f$.

**Lemma 0.1** (The linear approximation of the gradient uses the Hessian)**.** *Let $f$, $\underset{\sim}{x}$, the gradient $\underset{\sim}{g}$ and the Hessian $\mathbf{H}$ be as in the quadratic approximation, and let the second partial derivatives of $f$ be continuous at every point of an open ball around $\underset{\sim}{x}$. Then the linear approximation of the gradient of $f$ at $\underset{\sim}{x}$ is the gradient plus the Hessian times the step:*

$$\underset{\sim}{g}_q(\underset{\sim}{\delta}) = \underset{\sim}{g} + \mathbf{H}\underset{\sim}{\delta}. \tag{4}$$

Proof

*Proof.* For each $i$ and $j$, the $i$th entry of $\frac{\partial}{\partial \underset{\sim}{x}}\left(\frac{\partial}{\partial x_j} f(\underset{\sim}{x})\right)$ is $\frac{\partial}{\partial x_i}\left(\frac{\partial}{\partial x_j} f(\underset{\sim}{x})\right)$, which is $[\mathbf{H}]_{ij}$ by the definitions of the Hessian and of the derivative of a vector-valued function. So, for each $j$,

$$\begin{aligned} \left[\underset{\sim}{g}_q(\underset{\sim}{\delta})\right]_j &= \frac{\partial}{\partial x_j} f(\underset{\sim}{x}) + \sum_{i=1}^{p} [\mathbf{H}]_{ij}\,\delta_i && \text{(Equation 1, as a sum)} \\ &= [\underset{\sim}{g}]_j + \sum_{i=1}^{p} [\mathbf{H}]_{ij}\,\delta_i && \text{(definition of } \underset{\sim}{g}\text{)} \\ &= [\underset{\sim}{g}]_j + \sum_{i=1}^{p} [\mathbf{H}]_{ji}\,\delta_i && (\mathbf{H} \text{ is symmetric}) \\ &= [\underset{\sim}{g}]_j + [\mathbf{H}\underset{\sim}{\delta}]_j && \text{(matrix-vector product)} \\ &= [\underset{\sim}{g} + \mathbf{H}\underset{\sim}{\delta}]_j. && \text{(vector addition)} \end{aligned} \tag{5}$$

The third line uses the symmetry of the Hessian, whose hypothesis the lemma assumes. □

**Example 0.3** (A gradient that equals its linear approximation)**.** For $f(\underset{\sim}{x}) = \frac{1}{2}(x_1^2 + 3x_2^2)$, the second function of Example 0.2, the gradient is $\underset{\sim}{g} = (x_1,\ 3x_2)^\top$, and the Hessian is

$\begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$ at every point. By Lemma 0.1, the linear approximation of the gradient at $\underset{\sim}{x}$ is

$$\begin{aligned} \underset{\sim}{g}_q(\underset{\sim}{\delta}) &= \begin{bmatrix} x_1 \\ 3x_2 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} \delta_1 \\ \delta_2 \end{bmatrix} && \text{(Equation 4)} \\ &= \begin{bmatrix} x_1 \\ 3x_2 \end{bmatrix} + \begin{bmatrix} \delta_1 \\ 3\delta_2 \end{bmatrix} && \text{(matrix-vector product)} \\ &= \begin{bmatrix} x_1 + \delta_1 \\ 3x_2 + 3\delta_2 \end{bmatrix} && \text{(vector addition)} \\ &= \begin{bmatrix} x_1 + \delta_1 \\ 3(x_2 + \delta_2) \end{bmatrix}, && \text{(factor out 3)} \end{aligned} \tag{6}$$

which is the gradient of $f$ at $\underset{\sim}{x} + \underset{\sim}{\delta}$: the gradient of this $f$ is linear, so its linear approximation is exact.

**Theorem 0.1** (The Newton step is the root of the linear approximation of the gradient)**.** *Let $f$, $\underset{\sim}{x}$, $g$ and the Hessian $\mathbf{H}$ be as in Lemma 0.1, with $\mathbf{H}$ invertible. The only step $\underset{\sim}{\delta}$ at which the linear approximation of the gradient is zero is the Newton step:*

$$\underset{\sim}{g}_q(\underset{\sim}{\delta}) = \underset{\sim}{0}_{p\times 1} \iff \underset{\sim}{\delta} = -\mathbf{H}^{-1}\underset{\sim}{g}. \tag{7}$$

Proof

*Proof.* By Lemma 0.1, $\underset{\sim}{g}_q(\underset{\sim}{\delta}) = \underset{\sim}{0}_{p\times 1}$ exactly when

$$\begin{aligned} \underset{\sim}{0}_{p\times 1} &= \underset{\sim}{g} + \mathbf{H}\underset{\sim}{\delta} && \text{(Equation 4)} \\ -\underset{\sim}{g} &= \mathbf{H}\underset{\sim}{\delta} && \text{(subtract } \underset{\sim}{g} \text{ from both sides)} \\ -\mathbf{H}^{-1}\underset{\sim}{g} &= \underset{\sim}{\delta}. && \text{(multiply both sides by } \mathbf{H}^{-1} \text{ on the left)} \end{aligned} \tag{8}$$

Each line implies the next and, because $\mathbf{H}$ is invertible, the one before, so $\underset{\sim}{\delta} = -\mathbf{H}^{-1}\underset{\sim}{g}$ is the only solution. □

**Example 0.4** (The Newton step at $x = -2.5$ as a root)**.** At $x = -2.5$, the linear approximation of the gradient of the cubic $f(x) = \frac{1}{3}x^3 - 4x$ is $\ell(\delta) = 2.25 - 5\delta$ (Example 0.1). Setting it to 0,

$$\begin{aligned} 0 &= 2.25 - 5\delta && \text{(Equation 2)} \\ 5\delta &= 2.25 && \text{(add } 5\delta \text{ to both sides)} \\ \delta &= 0.45, && \text{(divide both sides by 5)} \end{aligned} \tag{9}$$

so the next iterate is $-2.5 + 0.45 = -2.05$, the point where the tangent line in Figure 2

crosses 0. The Newton's method iteration $x^{(t+1)} = \frac{1}{2}\left(x^{(t)} + 4/x^{(t)}\right)$ of Example 0.1 gives the same iterate. Here $H = [-5]$ is not positive definite, so Theorem 0.1 does not describe this step, but Theorem 0.1 does.

Remark (Two ways to read the Newton step)

*Remark* 0.1 (Two ways to read the Newton step). Theorem 0.1 reads the Newton step as the minimizer of the quadratic approximation, which needs the Hessian $\mathbf{H}$ to be positive definite. Theorem 0.1 reads it as the root of the linear approximation of the gradient, which needs only $\mathbf{H}$ invertible. Newton's method solves $\frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}) = \underset{\sim}{0}_{p\times 1}$ by replacing the gradient with its linear approximation at each iterate, so it looks for a stationary point of any kind, not only a minimizer.

## Convergence

**Definition 0.1** (Quadratic convergence)**.** A sequence $\underset{\sim}{x}^{(0)}, \underset{\sim}{x}^{(1)}, \ldots$ in $\mathbb{R}^p$ that converges to $\underset{\sim}{x}^*$ **converges quadratically** [4] if there is a constant $C$ with

$$\left\|\underset{\sim}{x}^{(t+1)} - \underset{\sim}{x}^*\right\| \le C\left\|\underset{\sim}{x}^{(t)} - \underset{\sim}{x}^*\right\|^2 \qquad \text{for every } t.$$

**Example 0.1** (Halving the error is not quadratic convergence)**.** $x^{(t)} = 2^{-t}$ converges to 0, but it does not converge quadratically: the bound would need $2^{-(t+1)} \le C\,(2^{-t})^2 = C\,2^{-2t}$, that is, $2^{t-1} \le C$, which fails once $t > 1 + \log_2 C$. Each step only halves the error.

**Example 0.2** (The error squares at each step)**.** For the iteration $x^{(t+1)} = \frac{1}{2}(x^{(t)} + 4/x^{(t)})$ of Example 0.1,

$$\begin{aligned} x^{(t+1)} - 2 &= \frac{1}{2}\left(x^{(t)} + \frac{4}{x^{(t)}}\right) - 2 && \text{(the iteration)} \\ &= \frac{(x^{(t)})^2 + 4 - 4x^{(t)}}{2x^{(t)}} && \text{(common denominator)} \\ &= \frac{(x^{(t)} - 2)^2}{2x^{(t)}}. && \text{(the numerator is a perfect square)} \end{aligned}$$

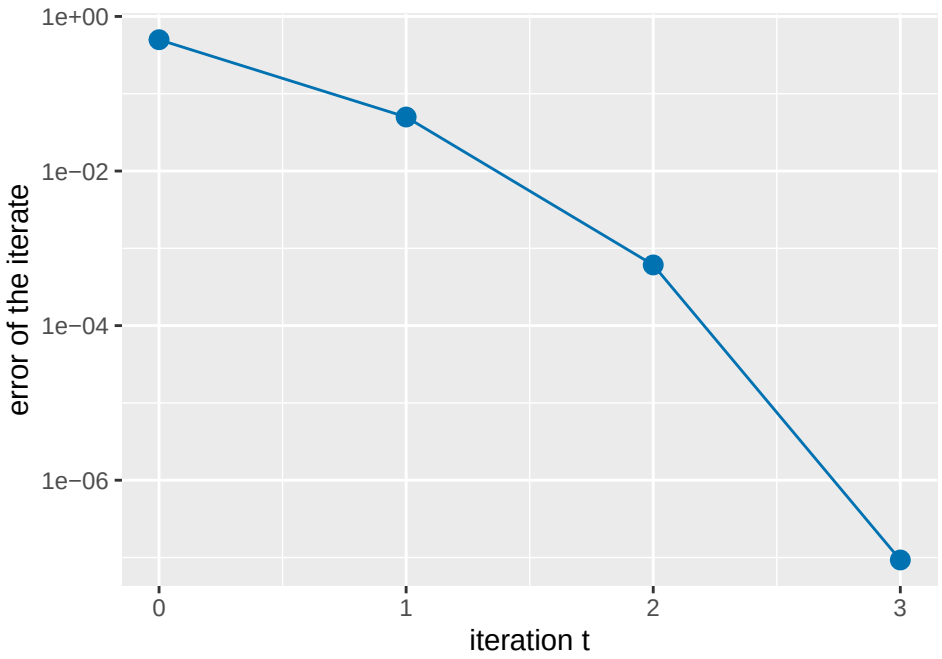
So near 2 the error is roughly squared and divided by 4 at each step: the errors are 0.05, then 0.00061, then 0.000000093, and $0.05^2/(2 \cdot 2.05) \approx 0.00061$. The iterates converge quadratically (Definition 0.1): $C = \frac{1}{2}$ works for every $x^{(t)} \ge 1$, since then $\frac{1}{2x^{(t)}} \le \frac{1}{2}$, and every iterate from 2.5 is at least 2, since the identity makes $x^{(t+1)} - 2 \ge 0$ whenever

$x^{(t)} > 0$. Zhou [1] states such a bound for Newton's method in general; it holds from starting points close enough to a minimizer where the Hessian is positive definite, when near that minimizer the Hessian changes by at most a constant times the distance moved, which is not proved here.

**Example 0.3** (Newton's method can find a maximum)**.** From $x^{(0)} = -2.5$ the same iteration gives $-2.05$, $-2.000610$, $-2.0000001$, approaching $-2$. But $-2$ is not a minimizer of $f$: it is a stationary point of $-f$, whose second derivative at $-2$ is $-f''(-2) = 4 > 0$, so it is a strict local minimizer of $-f$ (Theorem 7 in Critical Points and Optimality Conditions), that is, $f$ is strictly larger at $-2$ than at all nearby points. Newton's method looks only for a stationary point. In this example, the Hessian $H = f''(x) = 2x$ is negative at every iterate. If $H < 0$, then Theorem 0.1 does not apply, because it requires a positive definite Hessian. Instead, the quadratic approximation $q(\delta) = f + g\,\delta + \frac{1}{2}H\delta^2$ tends to $-\infty$ as $|\delta| \to \infty$, so it has no minimizer.
Figure 4 shows the cubic and its quadratic approximation.

**R**

```r
nc_x_nc <- numeric(4)
nc_x_nc[1] <- 2.5
for (nc_t_nc in 1:3) {
  nc_prev_nc <- nc_x_nc[nc_t_nc]
  nc_x_nc[nc_t_nc + 1] <- 0.5 * (nc_prev_nc + 4 / nc_prev_nc)
}
nc_steps_nc <- tibble::tibble(t = 0:3, error = nc_x_nc - 2)
ggplot2::ggplot(nc_steps_nc, ggplot2::aes(t, error)) +
  ggplot2::geom_line(colour = "#0072B2") +
  ggplot2::geom_point(colour = "#0072B2", size = 3) +
  ggplot2::scale_y_log10() +
  ggplot2::labs(x = "iteration t", y = "error of the iterate")
```




**Python**

```python
import matplotlib.pyplot as plt

nc_x_ax = [2.5]
for _ in range(3):
    nc_x_ax.append(0.5 * (nc_x_ax[-1] + 4 / nc_x_ax[-1]))
nc_err_ax = [x - 2 for x in nc_x_ax]
fig, ax = plt.subplots(figsize=(5, 3.5), layout="constrained")
ax.semilogy(range(4), nc_err_ax, "o-", color="#0072B2")
#> [<matplotlib.lines.Line2D object at 0x7fe1c9a0da90>]
ax.set_xlabel("iteration t")
#> Text(0.5, 0, 'iteration t')
ax.set_ylabel("error of the iterate")
#> Text(0, 0.5, 'error of the iterate')
ax.set_xticks(range(4))
#> [<matplotlib.axis.XTick object at 0x7fe1c99e74d0>, <matplotlib.axis.XTick object at 0x7fe
plt.show();
```

**R**

```
nx_x_nx <- seq(-4, -0.5, length.out = 301)
nx_f_nx <- nx_x_nx^3 / 3 - 4 * nx_x_nx
nx_f0_nx <- (-2.5)^3 / 3 - 4 * (-2.5)
nx_dx_nx <- nx_x_nx + 2.5
nx_q_nx <- nx_f0_nx + 2.25 * nx_dx_nx - 2.5 * nx_dx_nx^2
nx_dstep_nx <- -2.05 + 2.5
nx_q_step_nx <- nx_f0_nx + 2.25 * nx_dstep_nx - 2.5 * nx_dstep_nx^2
nx_f_max_nx <- (-2)^3 / 3 - 4 * (-2)
nx_curve_nx <- tibble::tibble(x = nx_x_nx, y = nx_f_nx)
nx_model_nx <- tibble::tibble(x = nx_x_nx, y = nx_q_nx)
nx_start_nx <- tibble::tibble(x = -2.5, y = nx_f0_nx)
nx_step_nx <- tibble::tibble(x = -2.05, y = nx_q_step_nx)
nx_max_nx <- tibble::tibble(x = -2, y = nx_f_max_nx)
nx_aes_nx <- ggplot2::aes(x, y)
nx_ylim_nx <- c(-2, 6.5)
nx_curve_nx |>
  ggplot2::ggplot() +
  nx_aes_nx +
  ggplot2::geom_line(colour = "#0072B2") +
  ggplot2::geom_line(
    data = nx_model_nx, colour = "#D55E00", linetype = "dashed"
  ) +
  ggplot2::geom_point(data = nx_start_nx, colour = "#D55E00", size = 3) +
```

**Theorem 0.1** (Newton's method minimizes a quadratic in one step)**.** *Let* $\mathbf{S}$ *be a* $p \times p$ *positive definite matrix,* $\underset{\sim}{c} \in \mathbb{R}^p$*, and* $f(\underset{\sim}{x}) = \frac{1}{2}\underset{\sim}{x}^\top \mathbf{S}\underset{\sim}{x} - \underset{\sim}{c} \cdot \underset{\sim}{x}$*. From any* $\underset{\sim}{x}^{(0)}$*, Newton's method (Definition 0.1) gives* $\underset{\sim}{x}^{(1)} = \mathbf{S}^{-1}\underset{\sim}{c}$*, the unique minimizer of* $f$*.*

Proof

*Proof.* Partial derivatives of a sum or of a constant multiple are the sum or multiple of the partial derivatives, so the derivative of a quadratic form applies to $\frac{1}{2}\underset{\sim}{x}^\top \mathbf{S}\underset{\sim}{x}$ and the derivative of a linear combination to $\underset{\sim}{c} \cdot \underset{\sim}{x}$:

$$\begin{aligned}\frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}) &= \tfrac{1}{2}(2\mathbf{S}\underset{\sim}{x}) - \underset{\sim}{c} \\ &= \mathbf{S}\underset{\sim}{x} - \underset{\sim}{c};\end{aligned}$$

by the Hessian of a quadratic form with $\frac{1}{2}\mathbf{S}$ in place of $\mathbf{S}$, and because the gradient $-\underset{\sim}{c}$ of the linear term is constant, $\mathbf{H}_f(\underset{\sim}{x}) = \mathbf{S}$, which is invertible. So

$$\begin{aligned}\underset{\sim}{x}^{(1)} &= \underset{\sim}{x}^{(0)} - \mathbf{S}^{-1}(\mathbf{S}\underset{\sim}{x}^{(0)} - \underset{\sim}{c}) && \text{(definition of Newton's method)} \\ &= \underset{\sim}{x}^{(0)} - \mathbf{S}^{-1}\mathbf{S}\underset{\sim}{x}^{(0)} + \mathbf{S}^{-1}\underset{\sim}{c} && \text{(distribute)} \\ &= \underset{\sim}{x}^{(0)} - \underset{\sim}{x}^{(0)} + \mathbf{S}^{-1}\underset{\sim}{c} && (\mathbf{S}^{-1}\mathbf{S} = \mathbf{I}_p) \\ &= \mathbf{S}^{-1}\underset{\sim}{c}. && \text{(cancel)}\end{aligned}$$

The gradient at $\mathbf{S}^{-1}\underset{\sim}{c}$ is $\mathbf{S}\mathbf{S}^{-1}\underset{\sim}{c} - \underset{\sim}{c} = \underset{\sim}{0}_{p\times 1}$, and $f$ is strictly convex (Theorem 6 in Convexity in Several Variables, part 2), so $\mathbf{S}^{-1}\underset{\sim}{c}$ is a global minimizer (Corollary 2 in Convexity in Several Variables). It is the only one: a strictly convex function has at most one global minimizer, since the midpoint of two distinct global minimizers, which have equal values, would have a strictly smaller value (Definition 4 in Convexity in Several Variables with $t = \frac{1}{2}$). □

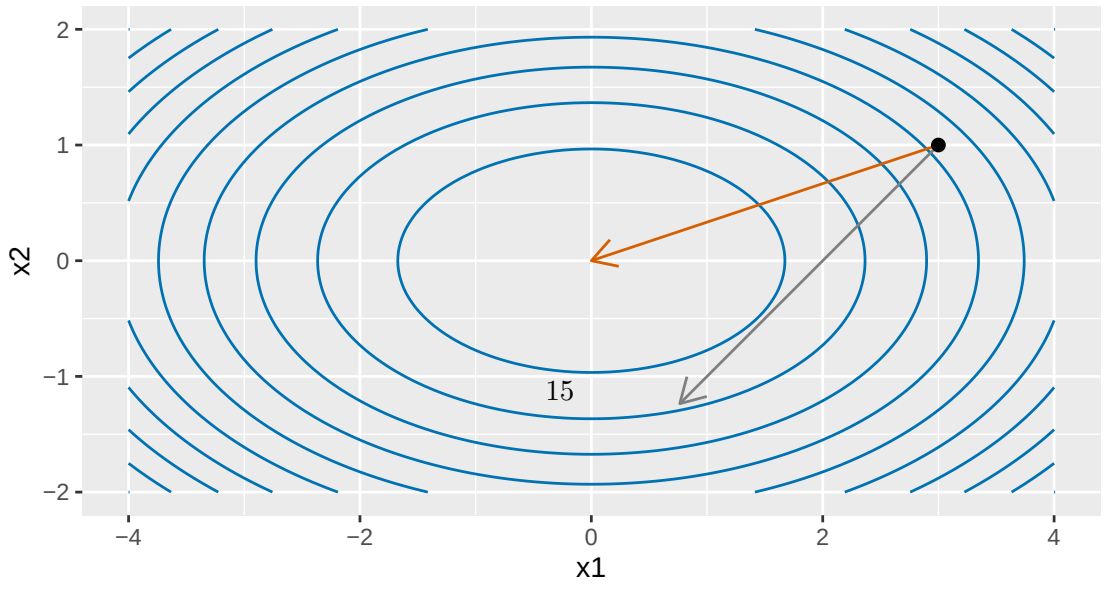
**Example 0.4** (One step to the minimizer)**.** Let $f(\underset{\sim}{x}) = \frac{1}{2}(x_1^2 + 3x_2^2)$, so $\mathbf{S} = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$ and $\underset{\sim}{c} = \underset{\sim}{0}_{2\times 1}$. From $\underset{\sim}{x}^{(0)} = (3,1)^\top$ the gradient is $\nabla f(\underset{\sim}{x}^{(0)}) = \mathbf{S}\underset{\sim}{x}^{(0)} = (3,3)^\top$, and $\mathbf{S}^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1/3 \end{bmatrix}$, so the Newton step is $-\mathbf{S}^{-1}(3,3)^\top = -(3,1)^\top$ and

$$\begin{aligned}\underset{\sim}{x}^{(1)} &= (3,1)^\top - (3,1)^\top \\ &= \underset{\sim}{0}_{2\times 1},\end{aligned}$$

the minimizer.
Figure 5 shows the step on the level curves of $f$.

**R**

```
nq_x1_nq <- seq(-4, 4, length.out = 101)
nq_x2_nq <- seq(-2, 2, length.out = 101)
nq_grid_nq <- tidyr::expand_grid(x1 = nq_x1_nq, x2 = nq_x2_nq)
nq_grid_nq$z <- (nq_grid_nq$x1^2 + 3 * nq_grid_nq$x2^2) / 2
nq_newton_nq <- tibble::tibble(x = 3, y = 1, xend = 0, yend = 0)
nq_grad_nq <- tibble::tibble(x = 3, y = 1, xend = 0.764, yend = -1.236)
nq_start_nq <- tibble::tibble(x1 = 3, x2 = 1)
nq_head_nq <- grid::unit(0.15, "inches")
nq_arrow_nq <- grid::arrow(length = nq_head_nq)
nq_aes_nq <- ggplot2::aes(x1, x2)
nq_z_aes_nq <- ggplot2::aes(z = z)
nq_seg_aes_nq <- ggplot2::aes(x = x, y = y, xend = xend, yend = yend)
nq_grid_nq |>
  ggplot2::ggplot() +
  nq_aes_nq +
  ggplot2::geom_contour(nq_z_aes_nq, colour = "#0072B2", bins = 10) +
  ggplot2::geom_segment(
    nq_seg_aes_nq,
    data = nq_grad_nq,
    colour = "grey50",
    arrow = nq_arrow_nq
  ) +
  ggplot2::geom_segment(
    nq_seg_aes_nq,
    data = nq_newton_nq,
    colour = "#D55E00",
    arrow = nq_arrow_nq
  ) +
  ggplot2::geom_point(data = nq_start_nq, size = 2) +
  ggplot2::coord_fixed() +
  ggplot2::labs(x = "x1", y = "x2")
```



> Figure 6 shows the same function as a surface.

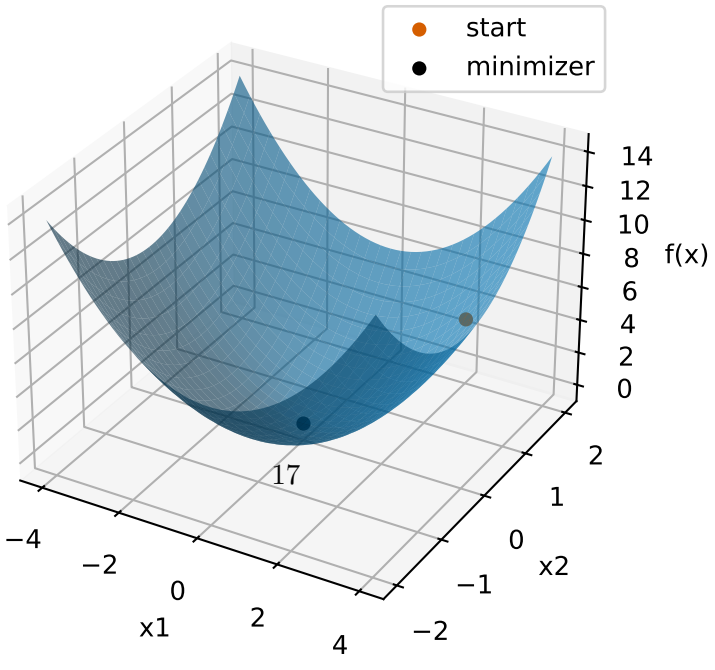
**R**

*(This 3D surface is interactive; see the HTML or slide version of this page to view it.)*

**Python**

```
import matplotlib.pyplot as plt
import numpy as np

nq_x1_3d, nq_x2_3d = np.meshgrid(np.linspace(-4, 4, 61),
                                 np.linspace(-2, 2, 61))
nq_z_3d = (nq_x1_3d**2 + 3 * nq_x2_3d**2) / 2
fig = plt.figure(figsize=(6, 4.5))
ax = fig.add_subplot(projection="3d")
ax.plot_surface(nq_x1_3d, nq_x2_3d, nq_z_3d, color="#0072B2", alpha=0.6)
#> <mpl_toolkits.mplot3d.art3d.Poly3DCollection object at 0x7fe1c9ac2900>
ax.scatter([3], [1], [6], color="#D55E00", label="start")
#> <mpl_toolkits.mplot3d.art3d.Path3DCollection object at 0x7fe1c9ac2a50>
ax.scatter([0], [0], [0], color="black", label="minimizer")
#> <mpl_toolkits.mplot3d.art3d.Path3DCollection object at 0x7fe1c7523250>
ax.set_xlabel("x1")
#> Text(0.5, 0, 'x1')
ax.set_ylabel("x2")
#> Text(0.5, 0.5, 'x2')
ax.set_zlabel("f(x)")
#> Text(0.5, 0, 'f(x)')
ax.legend(loc="upper right")
#> <matplotlib.legend.Legend object at 0x7fe1c9ac2ba0>
plt.show();
```



[1] H. Zhou, "Optimization and multivariate calculus." Lecture notes for Biostat 216, Mathematical Methods for Biostatistics, University of California, Los Angeles, 2024. Accessed: Oct. 05, 2026. [Online]. Available: https://ucla-biostat-216.github.io/2024fall/slides/13-optim/13-optim.html

[2] Wikipedia contributors, "Iterative method — Wikipedia, the free encyclopedia." 2026. Available: https://en.wikipedia.org/wiki/Iterative_method

[3] Wikipedia contributors, "Newton's method in optimization — Wikipedia, the free encyclopedia." 2026. Available: https://en.wikipedia.org/wiki/Newton%27s_method_in_optimization

[4] Wikipedia contributors, "Rate of convergence — Wikipedia, the free encyclopedia." 2026. Available: https://en.wikipedia.org/wiki/Rate_of_convergence